Predator reintroduction: annotated analysis v0.4

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1 Scope and changes from v0.3

This report extends `reintrov_0.3.Rmd` with two log-ratio evenness models, `m10` (extirpation/intact) and `m11` (reintroduction/extirpation). The first nine model formulas are retained. The new fits implement the evenness-model specification in the manuscript; they are newly reconstructed analyses, not recovered historical results.

Three manuscript figures are rebuilt: community responses to extirpation (four rows), predator responses to reintroduction (three rows), and community responses to reintroduction (four rows). Colours, scenario order, intact reference panels, dashed reference lines, and circle/square comparisons follow the Google Docs figures. Values are calculated from the fitted models, not traced from the old images. Explicit scenario values follow their captions.

Important corrections are documented rather than hidden:

- Predator biomass and stability are conditional on establishment. Failed predators within the inherited cohort remain in the binary establishment model. Zero biomass cannot be logged; v0.3 attempted to do so for some failures.
- "Short" and "Tall" now map to low and high trophic height respectively; v0.3's prediction table reversed those labels.
- Figure reference values (50 species, biomass 100, stability 0.1 and evenness 0.1) are illustrative, not measured ecosystem averages or physical units. Inverse-CV stability 0.1 means CV = 10, not 10%.
- For Figure 3, uncertainty in both sequential transitions is propagated with paired resampling of simulation records. Multiplying two separate interval endpoints, as in v0.3, does not give a calibrated interval for their product.
- These confidence intervals describe conditional model predictions within the simulated ensemble. They are not prediction intervals for real releases.

2 Input, phases and analysis population

stage	simulations
Input	69895
Finite predator biomass ratio	66959
Baseline: no imposed feeding-link change	10969

Each row is one simulation trajectory across three phases: intact (1), extirpated (2) and reintroduced (3). Suffixes `21` and `32` denote ratios between consecutive phases; `31` compares reintroduced with intact. Evenness ratios can exceed one, so they are analysed with log-normal linear models, not beta regression. The data contain evenness measurements but not saved historical evenness-model fits. `DATA_DICTIONARY.csv` in the repository describes the input columns.

For comparability, v0.4 retains v0.3's global finite-biomass-ratio filter. Before that filter there are 11614 baseline records; 645 are excluded by it. Consequently, the binary establishment model is conditional on the retained analysis cohort, not every attempted simulation. This selection and missing simulation outputs should be considered when interpreting establishment rates.

3 Eleven statistical models

	model	family
m1	m1	binomial logit
m2	m2	Gaussian on log ratio
m3	m3	Gaussian on log ratio
m4	m4	beta logit
m5	m5	beta logit
m6	m6	Gaussian on log ratio
m7	m7	Gaussian on log ratio
m8	m8	Gaussian on log ratio
m9	m9	Gaussian on log ratio
m10	m10	Gaussian on log ratio
m11	m11	Gaussian on log ratio

```
#> Warning in family$saturated.ll(y, prior.weights, theta): saturated
#> likelihood may be inaccurate

#> Warning in family$saturated.ll(y, prior.weights, theta): saturated
#> likelihood may be inaccurate

#> <environment: 0x108c81d28>
```

	model	family	n	excluded_from_baseline
m1	m1	binomial logit	10969	0
m2	m2	Gaussian on log ratio	10607	362
m3	m3	Gaussian on log ratio	10602	367
m4	m4	beta logit	10969	0
m5	m5	beta logit	10969	0
m6	m6	Gaussian on log ratio	10969	0
m7	m7	Gaussian on log ratio	10969	0
m8	m8	Gaussian on log ratio	10969	0
m9	m9	Gaussian on log ratio	10969	0
m10	m10	Gaussian on log ratio	10969	0
m11	m11	Gaussian on log ratio	10969	0

`m1` predicts establishment; `m2` and `m3` predict predator stability and biomass conditional on establishment. Pairs `m4/m5`, `m6/m7`, `m8/m9` and `m10/m11` describe resident richness, total biomass, community stability and evenness respectively. These are associations within the simulated ensemble. In particular, conditioning on predator survival in community models does not estimate a randomised causal effect of survival.

The richness beta regressions retain v0.3's `mgcv::betar` specification, including its default endpoint adjustment: exact zero/one proportions are moved just inside the unit interval (`eps = 100 * .Machine$double.eps`). There are 5809 exact-one extirpation ratios and 8835 exact-one reintroduction ratios. This substantial boundary mass is a limitation of the inherited beta models; a boundary-aware alternative would be a separate sensitivity analysis, not a silent change to the manuscript's established specification.

4 Explicit manuscript scenarios

Factor	Left	Value	Right	Value
Community connectance	Strong	0.09	Weak	0.06
Community intactness	Intact	1.00	Depleted	0.50
Food-web trophic height	Short	2.05	Tall	3.43
Mass ratio Predator:Prey	Large	1000.00	Small	10.00
Predator diet breadth	Large	12.00	Small	1.00
Predator survival	High	1.00	Low	0.00
Prey intactness	Intact	1.00	Depleted	0.50

These profiles hold unspecified predictors fixed; they do not represent observed whole communities. Reference log-stability is the mean of logged stability. Prey intactness, community intactness and survival can therefore be varied as conditional covariates without simulating a new community. Some settings are unusual combinations; predictions are not empirical recommendations about optimal release conditions. If a scenario variable is absent from a model, changing it leaves that model's prediction unchanged. For example, the predator-survival setting is not a predictor in the population models; its identical Figure 2 points are retained to follow the manuscript's seven-column layout. Each regression is evaluated separately. A prescribed intactness value in a second-transition model need not equal the first-transition richness prediction at the reference profile. Consequently, products of these predictions are illustrative model comparisons, not reconstructions of a single internally constrained food-web trajectory.

5 Sequential community responses and uncertainty

The paired bootstrap uses 200 successful resamples out of 200, with seed 20261103. It resamples whole simulation records, not time steps. It assumes the retained simulation records can be treated as independent sampling units. It does not include uncertainty in the chosen reference profile or in the underlying food-web parameterisation. More replicates can be requested through the `bootstrap_replicates` parameter.

200 resamples generated numerical warnings, recorded in `bootstrap_status.csv`. In particular, saturated-likelihood warnings from the inherited beta family must be considered alongside the large mass at persistence = 1; convergence alone does not establish that this distributional specification is appropriate.

warning	replicates
saturated likelihood may be inaccurate	200

6 Rebuilt manuscript figures

6.1 Figure 1: community responses to extirpation

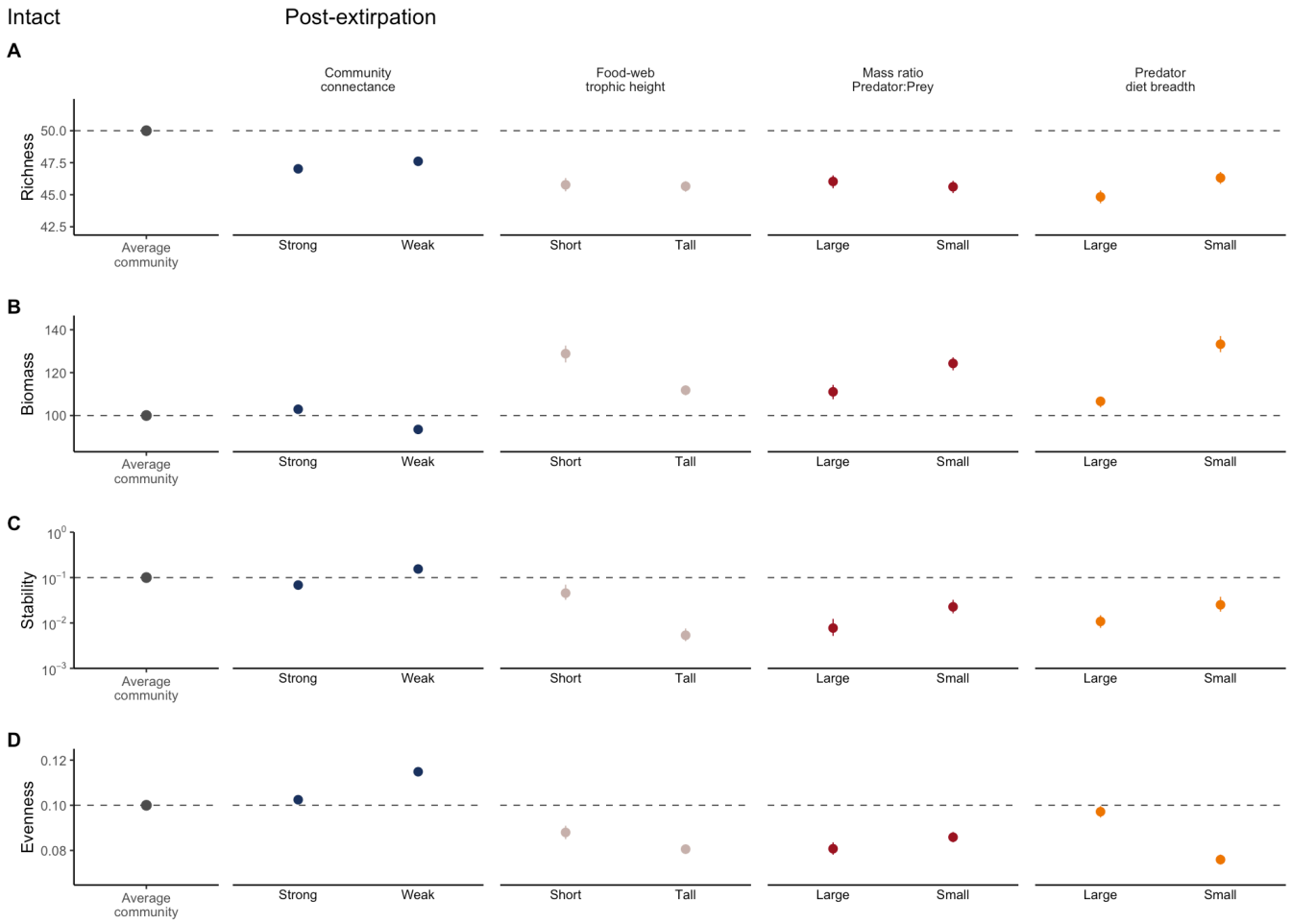


Figure 1: Conditional community responses to predator extirpation. Four metrics and four scenario pairs follow the manuscript layout. Dots are model predictions; intervals are 95% paired-bootstrap intervals. Dashed lines show illustrative intact reference values. Stability uses a logarithmic axis.

6.2 Figure 2: predator responses to reintroduction

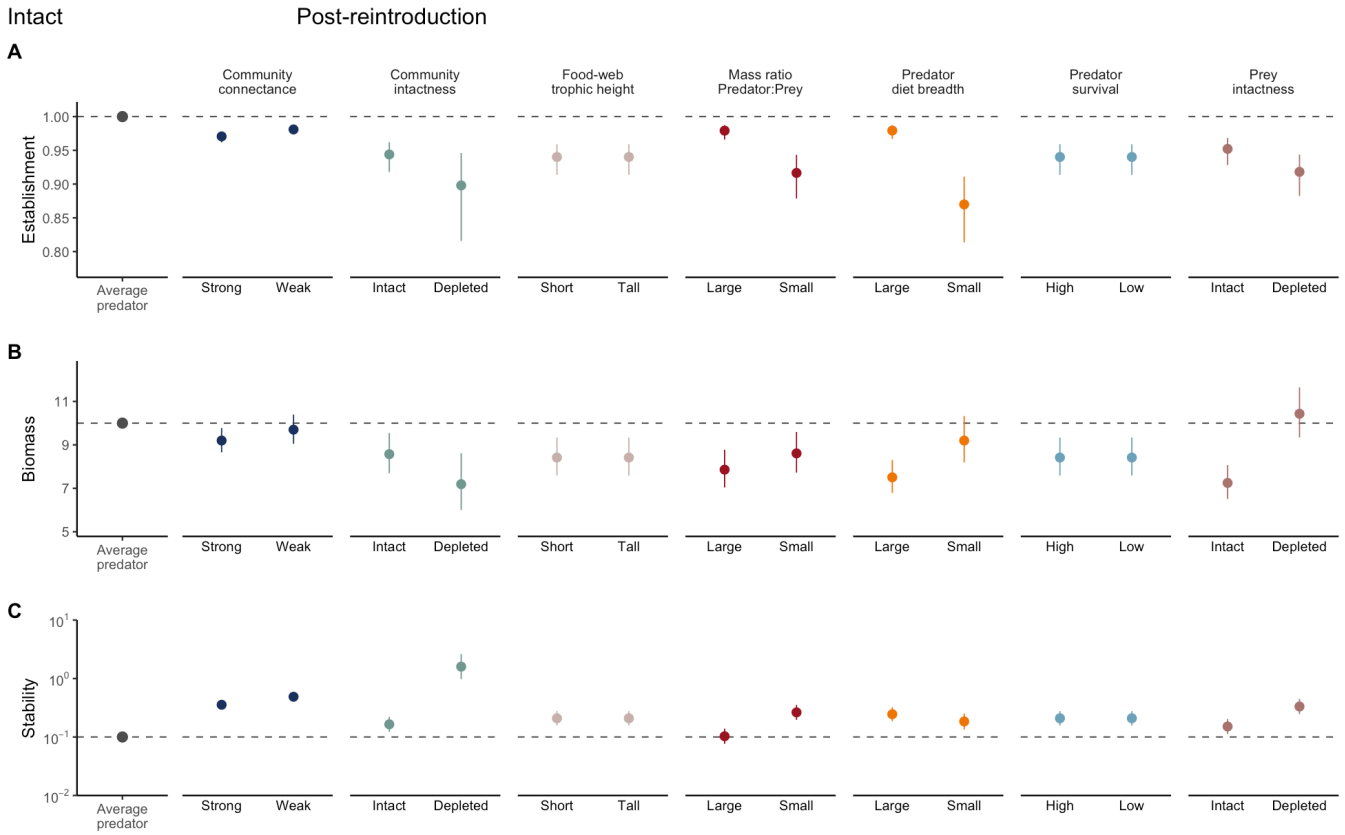


Figure 2: Predator establishment probability, biomass and temporal stability. Biomass and stability predictions are conditional on predator establishment. Intervals are pointwise 95% model confidence intervals, transformed from the link/log scale. The population reference values are probability 1, biomass 10 and inverse-CV stability 0.1.

6.3 Figure 3: community responses after reintroduction

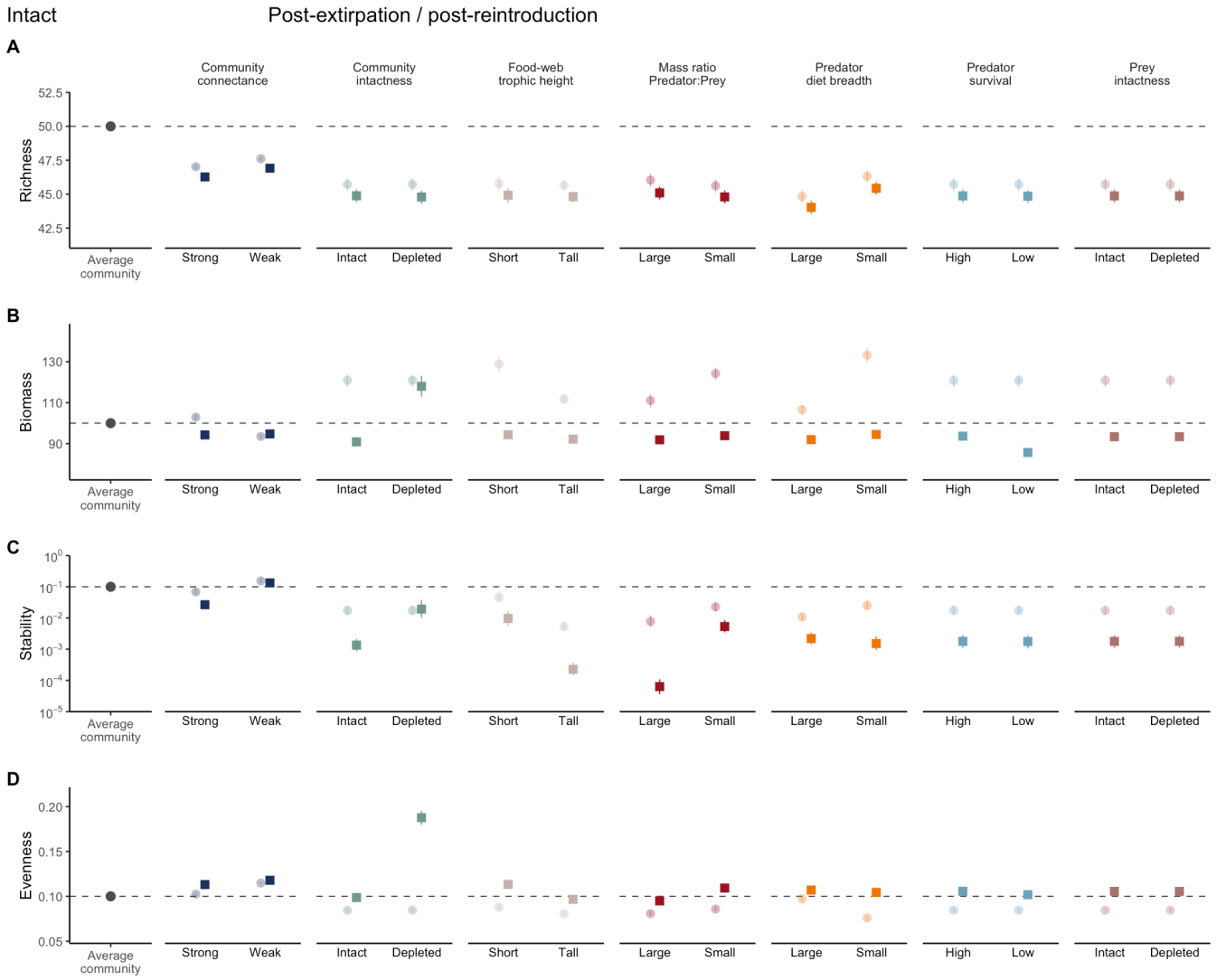
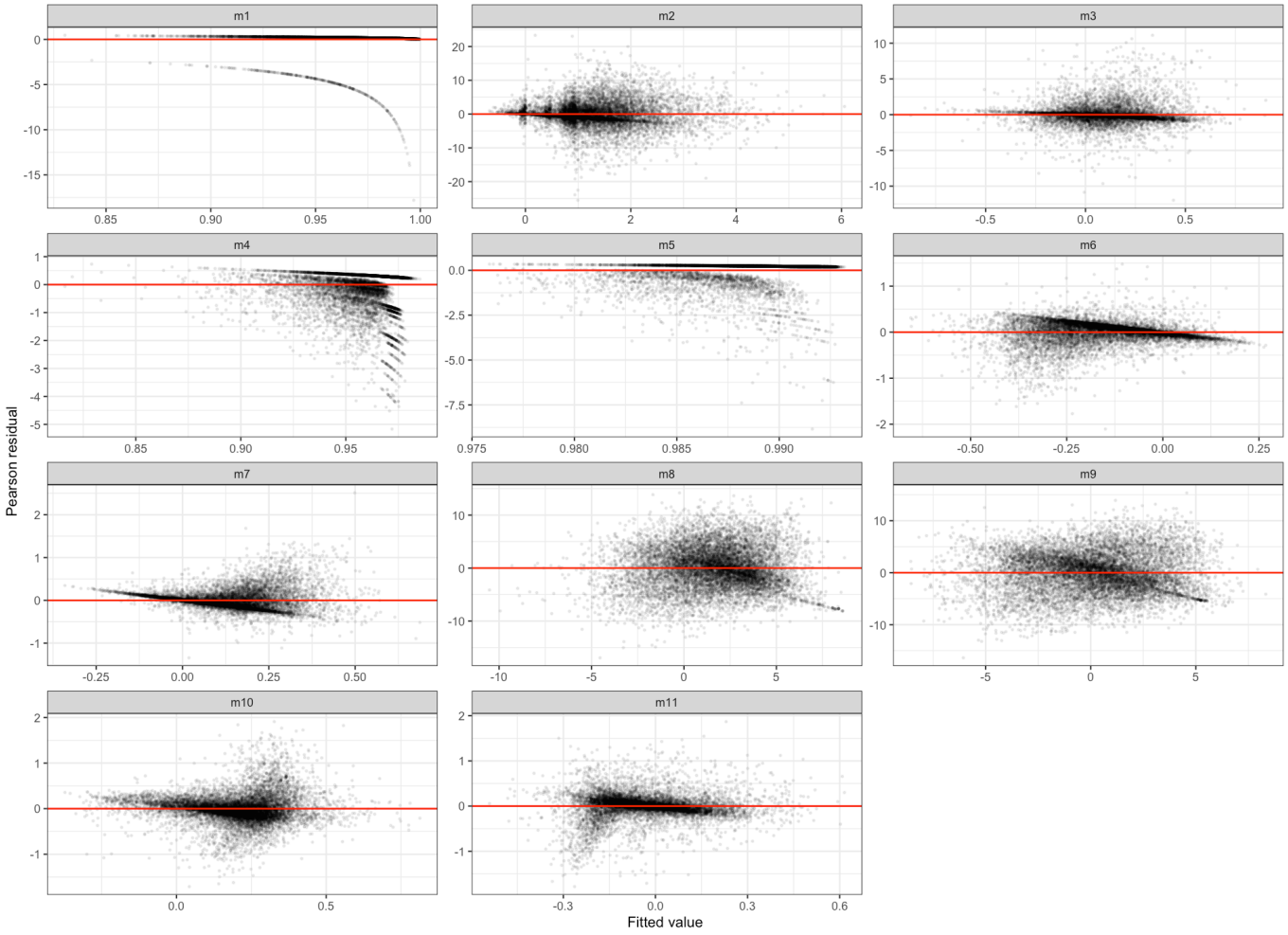


Figure 3: Community responses on a common intact-reference scale. Transparent circles show post-extirpation predictions and opaque squares show post-reintroduction predictions. Reintroduction values combine both sequential model ratios. Intervals propagate their dependence using the paired bootstrap. Evenness is newly reconstructed in v0.4.

7 Model diagnostics and detailed results

Model	n	AIC	Residual SD	Max residual	Converged
m1	10969	3037.1941	1.0172	17.8051	TRUE
m2	10607	56950.1770	3.5421	25.5156	TRUE
m3	10602	35703.6031	1.3021	11.9756	TRUE
m4	10969	-306890.0889	0.6657	5.1658	TRUE
m5	10969	-475997.5872	0.6530	8.8436	TRUE
m6	10969	-619.7618	0.2350	2.1044	TRUE
m7	10969	-1888.1941	0.2218	2.5108	TRUE
m8	10969	60850.3983	3.8730	16.9453	TRUE
m9	10969	60657.0233	3.8383	16.3696	TRUE
m10	10969	5418.5103	0.3095	1.9087	TRUE
m11	10969	1695.5238	0.2612	1.8680	TRUE



7.1 m1

Formula:

```
 delta_sp_32 ~ realised_C_1 * realised_S_1 + g_1 * delta_g_31 +
delta_s_21 + z_1
```

```
Call:
glm(formula = model_formulas[[id]], family = binomial(link = "logit"),
    data = data)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-3.3947   0.1720   0.2369   0.2999   0.6097

Coefficients:
                            Estimate Std. Error z value Pr(>|z|)
(Intercept)                1.3263038  0.7592107   1.747 0.080646 .
realised_C_1              -0.1477199  1.5178120  -0.097 0.922469
realised_S_1               0.0137905  0.0094220   1.464 0.143289
g_1                        0.0516643  0.0485779   1.064 0.287539
delta_g_31                 0.2239613  0.3624661   0.618 0.536653
delta_s_21                 1.2951460  0.6750597   1.919 0.055039 .
z_1                        0.0014682  0.0002306   6.366 1.94e-10 ***
realised_C_1:realised_S_1 -0.2923196  0.0786419  -3.717 0.000202 ***
g_1:delta_g_31             0.1588710  0.0680804   2.334 0.019618 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 3174.9  on 10968  degrees of freedom
Residual deviance: 3019.2  on 10960  degrees of freedom
AIC: 3037.2

Number of Fisher Scoring iterations: 7
```

7.2 m2

Formula:

```
 log(delta_stabp_31) ~ realised_C_1 * realised_S_1 + g_1 * delta_g_31 +
delta_s_21 + z_1
```

```
Call:
lm(formula = model_formulas[[id]], data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-25.5156  -1.4809  -0.3306   1.4118  23.3318

Coefficients:
                            Estimate Std. Error t value Pr(>|t|)
(Intercept)                6.2239239  0.5381725  11.565  < 2e-16 ***
realised_C_1               1.6978738  0.9150960   1.855 0.063566 .
realised_S_1               0.0219007  0.0056689   3.863 0.000113 ***
g_1                        0.1200170  0.0266358   4.506 6.68e-06 ***
delta_g_31                -0.8878407  0.2498271  -3.554 0.000381 ***
delta_s_21                -4.5417159  0.4968707  -9.141  < 2e-16 ***
z_1                       -0.0009494  0.0001077  -8.815  < 2e-16 ***
```

```
realised_C_1:realised_S_1 -0.2449197  0.0511673  -4.787 1.72e-06 ***
g_1:delta_g_31            -0.1183862  0.0353658  -3.347 0.000818 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.543 on 10598 degrees of freedom
Multiple R-squared:  0.04841,   Adjusted R-squared:  0.04769
F-statistic: 67.39 on 8 and 10598 DF,  p-value: < 2.2e-16
```

7.3 m3

Formula:

```
 log(delta_bmp_31) ~ realised_C_1 * realised_S_1 + g_1 * delta_g_31 +
delta_s_21 + z_1


Call:
lm(formula = model_formulas[[id]], data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-11.9756  -0.3904  -0.0878   0.1506  11.1059

Coefficients:
                            Estimate Std. Error t value Pr(>|t|)
(Intercept)                1.810e-01  1.979e-01   0.915 0.360441
realised_C_1               1.156e+00  3.365e-01   3.434 0.000597 ***
realised_S_1               5.643e-04  2.087e-03   0.270 0.786815
g_1                        2.327e-02  9.792e-03   2.377 0.017489 *
delta_g_31                -4.233e-01  9.187e-02  -4.607 4.12e-06 ***
delta_s_21                 3.517e-01  1.827e-01   1.925 0.054212 .
z_1                       -9.206e-05  3.960e-05  -2.325 0.020107 *
realised_C_1:realised_S_1 -5.839e-02  1.882e-02  -3.103 0.001923 **
g_1:delta_g_31            -5.255e-02  1.300e-02  -4.041 5.35e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.303 on 10593 degrees of freedom
Multiple R-squared:  0.01855,   Adjusted R-squared:  0.01781
F-statistic: 25.03 on 8 and 10593 DF,  p-value: < 2.2e-16
```

7.4 m4

Formula:

```
 delta_s_21 ~ realised_C_1 * realised_S_1 + g_1 + z_1 + stab_com_1 +
tlvl_mean_1


Family: Beta regression(1.609)
Link function: logit
```

```
Formula:
delta_s_21 ~ realised_C_1 * realised_S_1 + g_1 + z_1 + stab_com_1 +
    tlvl_mean_1

Parametric coefficients:
                           Estimate Std. Error z value Pr(>|z|)
(Intercept)                3.459e+00  8.575e-02  40.341  < 2e-16 ***
realised_C_1               2.016e+00  2.592e-01   7.777 7.45e-15 ***
realised_S_1               1.114e-03  1.644e-03   0.677  0.49816
g_1                       -3.375e-02  2.370e-03 -14.241  < 2e-16 ***
z_1                        1.095e-04  3.461e-05   3.163  0.00156 **
stab_com_1                 1.267e-02  2.479e-03   5.112 3.19e-07 ***
tlvl_mean_1               -2.133e-02  2.490e-02  -0.857  0.39171
realised_C_1:realised_S_1 -1.951e-01  1.610e-02 -12.115  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


R-sq.(adj) =  0.0476   Deviance explained = -1.55%
-REML = -1.5342e+05  Scale est. = 1         n = 10969
```

7.5 m5

Formula:

```
 delta_s_32 ~ realised_C_1 * realised_S_1 + delta_s_21 + g_1 + z_1 +
tlvl_mean_1 + delta_sp_32 + stab_com_2



Family: Beta regression(3.837)
Link function: logit

Formula:
delta_s_32 ~ realised_C_1 * realised_S_1 + delta_s_21 + g_1 +
    z_1 + tlvl_mean_1 + delta_sp_32 + stab_com_2

Parametric coefficients:
                           Estimate Std. Error z value Pr(>|z|)
(Intercept)                4.440e+00  1.572e-01  28.246  < 2e-16 ***
realised_C_1              -1.335e-02  2.541e-01  -0.053 0.958119
realised_S_1              -9.222e-03  1.598e-03  -5.771 7.86e-09 ***
delta_s_21                 2.146e-01  1.270e-01   1.690 0.091065 .
g_1                        6.161e-03  2.313e-03   2.664 0.007718 **
z_1                       -1.253e-04  3.302e-05  -3.793 0.000149 ***
tlvl_mean_1                4.980e-05  2.442e-02   0.002 0.998373
delta_sp_32                3.043e-02  5.439e-02   0.560 0.575777
stab_com_2                 1.193e-02  2.281e-03   5.230 1.70e-07 ***
realised_C_1:realised_S_1 -6.031e-02  1.566e-02  -3.852 0.000117 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1


R-sq.(adj) =  0.0363   Deviance explained = -0.24%
-REML = -2.3797e+05  Scale est. = 1         n = 10969
```

7.6 m6

Formula:

```
 log(delta_bmc_21) ~ realised_C_1 * realised_S_1 + g_1 + z_1 +
stab_com_1 + tlvl_mean_1
```

```
Call:
lm(formula = model_formulas[[id]], data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-2.10440 -0.08519  0.03050  0.13175  1.47894

Coefficients:
                           Estimate Std. Error t value Pr(>|t|)
(Intercept)                2.956e-01  1.948e-02  15.170  < 2e-16 ***
realised_C_1              -1.863e-01  5.912e-02  -3.151  0.00163 **
realised_S_1               1.166e-03  3.724e-04   3.132  0.00174 **
g_1                       -2.026e-02  5.294e-04 -38.265  < 2e-16 ***
z_1                       -1.138e-04  7.862e-06 -14.471  < 2e-16 ***
stab_com_1                -1.457e-02  5.644e-04 -25.825  < 2e-16 ***
tlvl_mean_1               -1.028e-01  5.636e-03 -18.243  < 2e-16 ***
realised_C_1:realised_S_1  6.759e-02  3.630e-03  18.620  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2351 on 10961 degrees of freedom
Multiple R-squared:  0.2454,    Adjusted R-squared:  0.2449
F-statistic: 509.1 on 7 and 10961 DF,  p-value: < 2.2e-16
```

7.7 m7

Formula:

```
 log(delta_bmc_32) ~ realised_C_1 * realised_S_1 + delta_s_21 + g_1 +
z_1 + tlvl_mean_1 + delta_sp_32 + stab_com_2
```

```
Call:
lm(formula = model_formulas[[id]], data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-1.34284 -0.12480 -0.03076  0.07787  2.51077

Coefficients:
                           Estimate Std. Error t value Pr(>|t|)
(Intercept)                3.256e-01  3.454e-02   9.425  < 2e-16 ***
realised_C_1               3.806e-01  5.589e-02   6.811 1.02e-11 ***
realised_S_1              -1.637e-03  3.508e-04  -4.667 3.09e-06 ***
delta_s_21                -5.224e-01  2.791e-02 -18.718  < 2e-16 ***
g_1                        1.777e-02  5.080e-04  34.989  < 2e-16 ***
z_1                        9.188e-05  7.252e-06  12.669  < 2e-16 ***
```

```
tlvl_mean_1                8.595e-02  5.365e-03  16.022  < 2e-16 ***
delta_sp_32                8.905e-02  1.195e-02   7.452 9.91e-14 ***
stab_com_2                 1.689e-03  5.016e-04   3.368  0.00076 ***
realised_C_1:realised_S_1 -7.511e-02  3.437e-03 -21.855  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2219 on 10959 degrees of freedom
Multiple R-squared:  0.2361,    Adjusted R-squared:  0.2355
F-statistic: 376.4 on 9 and 10959 DF,  p-value: < 2.2e-16
```

7.8 m8

Formula:

```
 log(delta_stabc_21) ~ realised_C_1 * realised_S_1 + g_1 + z_1 +
stab_com_1 + tlvl_mean_1



Call:
lm(formula = model_formulas[[id]], data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-16.9453  -2.4566  -0.0581   2.4704  14.1978

Coefficients:
                           Estimate Std. Error t value Pr(>|t|)
(Intercept)               15.0696960  0.3210290  46.942  < 2e-16 ***
realised_C_1               5.6412348  0.9742157   5.791 7.21e-09 ***
realised_S_1              -0.0138951  0.0061355  -2.265   0.0235 *
g_1                       -0.0766990  0.0087234  -8.792  < 2e-16 ***
z_1                       -0.0010868  0.0001296  -8.389  < 2e-16 ***
stab_com_1                -0.5672360  0.0092991 -60.999  < 2e-16 ***
tlvl_mean_1               -1.5498226  0.0928679 -16.688  < 2e-16 ***
realised_C_1:realised_S_1 -0.6563143  0.0598165 -10.972  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.874 on 10961 degrees of freedom
Multiple R-squared:  0.2762,    Adjusted R-squared:  0.2757
F-statistic: 597.4 on 7 and 10961 DF,  p-value: < 2.2e-16
```

7.9 m9

Formula:

```
 log(delta_stabc_32) ~ realised_C_1 * realised_S_1 + delta_s_21 + g_1 +
z_1 + tlvl_mean_1 + delta_sp_32 + stab_com_2



Call:
lm(formula = model_formulas[[id]], data = data)
```

```
Residuals:
     Min       1Q   Median       3Q      Max
-16.3696  -2.5764   0.0764   2.5542  15.3545

Coefficients:
                            Estimate Std. Error t value Pr(>|t|)
(Intercept)                18.0007077  0.5978007  30.112   <2e-16 ***
realised_C_1               -1.4900021  0.9670935  -1.541    0.123
realised_S_1               -0.0064243  0.0060707  -1.058    0.290
delta_s_21                 -5.3125451  0.4830032 -10.999   <2e-16 ***
g_1                         0.1103430  0.0087904  12.553   <2e-16 ***
z_1                        -0.0033898  0.0001255 -27.010   <2e-16 ***
tlvl_mean_1                -1.1711224  0.0928349 -12.615   <2e-16 ***
delta_sp_32                 0.0125349  0.2068073   0.061    0.952
stab_com_2                 -0.5340021  0.0086805 -61.518   <2e-16 ***
realised_C_1:realised_S_1 -0.4996927  0.0594779  -8.401   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.84 on 10959 degrees of freedom
Multiple R-squared:  0.3135,    Adjusted R-squared:  0.3129
F-statistic:   556 on 9 and 10959 DF,  p-value: < 2.2e-16
```

7.10 m10

Formula:

```
 log(delta_ev_21) ~ realised_C_1 * realised_S_1 + g_1 + z_1 + stab_com_1
+ tlvl_mean_1


Call:
lm(formula = model_formulas[[id]], data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-1.71016 -0.14764 -0.01512  0.13125  1.90873

Coefficients:
                            Estimate Std. Error t value Pr(>|t|)
(Intercept)                4.676e-01  2.566e-02  18.226  < 2e-16 ***
realised_C_1               2.105e-01  7.786e-02   2.704  0.00686 **
realised_S_1              -8.799e-05  4.903e-04  -0.179  0.85759
g_1                        2.241e-02  6.972e-04  32.151  < 2e-16 ***
z_1                       -6.192e-05  1.035e-05  -5.980 2.30e-09 ***
stab_com_1                -3.080e-03  7.432e-04  -4.145 3.42e-05 ***
tlvl_mean_1               -6.374e-02  7.422e-03  -8.588  < 2e-16 ***
realised_C_1:realised_S_1 -8.035e-02  4.780e-03 -16.808  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3096 on 10961 degrees of freedom
Multiple R-squared:  0.1695,    Adjusted R-squared:  0.169
F-statistic: 319.6 on 7 and 10961 DF,  p-value: < 2.2e-16
```

7.11 m11

Formula:

```
 log(delta_ev_32) ~ realised_C_1 * realised_S_1 + delta_s_21 + g_1 + z_1
+ tlvl_mean_1 + delta_sp_32 + stab_com_2
```

```
Call:
lm(formula = model_formulas[[id]], data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-1.78219 -0.08044  0.00533  0.09671  1.86799

Coefficients:
                           Estimate Std. Error t value Pr(>|t|)
(Intercept)                1.389e+00  4.068e-02  34.158  < 2e-16 ***
realised_C_1              -3.323e-01  6.580e-02  -5.050 4.49e-07 ***
realised_S_1               9.253e-05  4.131e-04   0.224  0.82274
delta_s_21                -1.286e+00  3.286e-02 -39.121  < 2e-16 ***
g_1                       -2.015e-02  5.981e-04 -33.692  < 2e-16 ***
z_1                       -7.981e-05  8.539e-06  -9.347  < 2e-16 ***
tlvl_mean_1               -5.088e-02  6.317e-03  -8.056 8.72e-16 ***
delta_sp_32                3.643e-02  1.407e-02   2.589  0.00964 **
stab_com_2                -3.889e-03  5.906e-04  -6.584 4.80e-11 ***
realised_C_1:realised_S_1  5.525e-02  4.047e-03  13.652  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2613 on 10959 degrees of freedom
Multiple R-squared:  0.2083,    Adjusted R-squared:  0.2077
F-statistic: 320.5 on 9 and 10959 DF,  p-value: < 2.2e-16
```

8 Comparison with v0.3 and the manuscript images

The v0.3 formulas for `m1` to `m9` are preserved; the populations for predator continuous responses are now explicitly restricted as described above. The figure settings are deliberately explicit rather than recomputing caption values from quantiles that may change with filtering. In particular, trophic height labels are corrected and the high/low survival settings are 1 and 0. All other predictors remain fixed at the exported reference profile.

The new evenness fits and the uncertainty correction can change points and intervals. The Google Docs images should be replaced only after comparing these outputs and revisiting any numerical statements that depend on them. In particular, correcting Short/Tall changes the interpretation of the trophic-height panels, and conditioning predator biomass on establishment can change relationships with prey persistence. Reconstructed outputs should not be described as exact numerical reproductions of the historical images. This report does not silently replace the manuscript figures or rewrite its results. Re-establishing the source revision that generated the historical simulation data remains a separate provenance question.

9 Reproducibility

Run from the repository root, with `sim.Rdata` beside this file:

```r
rmarkdown::render("reintrov_0.4.Rmd", output_format = "pdf_document")
# HTML provides a searchable report; keep_md writes the executed Markdown too.
rmarkdown::render("reintrov_0.4.Rmd", output_format = "html_document",
                  output_options = list(keep_md = TRUE))
```

PDF rendering needs Pandoc and a LaTeX installation (for example TinyTeX). HTML rendering needs Pandoc but not LaTeX. The source shows all fitting and plotting code. The results directory contains the eleven fitted models, model sample sizes, coefficients, scenarios, predictions, diagnostics, bootstrap status, and standalone figure PDFs/PNGs. Packages and versions are reported below. No package installation or external data download is performed.

```
#> R version 4.2.3 (2023-03-15)
#> Platform: aarch64-apple-darwin20 (64-bit)
#> Running under: macOS 26.5.1
#>
#> Matrix products: default
#> BLAS:   /Library/Frameworks/R.framework/Versions/4.2-arm64/Resources/lib/libRblas.0.dylib
#> LAPACK: /Library/Frameworks/R.framework/Versions/4.2-arm64/Resources/lib/libRlapack.dylib
#>
#> locale:
#> [1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#> [1] patchwork_1.2.0 ggplot2_3.5.1
#>
#> loaded via a namespace (and not attached):
#>  [1] knitr_1.45         magrittr_2.0.3     splines_4.2.3
#>  [4] tidyselect_1.2.0   lattice_0.22-5     R6_2.5.1
#>  [7] ragg_1.2.7         rlang_1.1.3        fastmap_1.1.1
#> [10] fansi_1.0.6        dplyr_1.1.4        tools_4.2.3
#> [13] grid_4.2.3         nlme_3.1-164       gtable_0.3.6
#> [16] mgcv_1.9-1         xfun_0.42          utf8_1.2.4
#> [19] cli_3.6.2          withr_3.0.1        systemfonts_1.0.5
#> [22] htmltools_0.5.7    yaml_2.3.12        digest_0.6.34
#> [25] tibble_3.2.1       lifecycle_1.0.4    textshaping_0.3.7
#> [28] Matrix_1.6-5       RColorBrewer_1.1-3 farver_2.1.1
#> [31] vctrs_0.6.5        glue_1.7.0         evaluate_0.23
#> [34] rmarkdown_2.25     labeling_0.4.3     compiler_4.2.3
#> [37] pillar_1.9.0       generics_0.1.3     scales_1.4.0
#> [40] pkgconfig_2.0.3
```